LESSON 3.3
MORE WITH POWERS OF 10

LESSON INTRODUCTION

MA.8.NSO.1.3 Extend previous understanding of the laws of exponents to include integer exponents. Apply the laws of exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to integer exponents and rational number bases, with procedural fluency.

LEARNING TARGETS

- I can use laws of exponents to evaluate numerical expressions with a base of 10 and exponents less than 0.

BENCHMARK COHERENCE

BUILDING ON

MA.7.NSO.1.1 Know and apply the laws of exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to whole-number exponents and rational number bases.

WORKING TOWARD

MA.912.NSO.1.1 Extend previous understanding of the laws of exponents to include rational number exponents. Apply the laws of exponents to evaluate numerical expressions and generate equivalent numerical expressions involving rational exponents.

ESSENTIAL QUESTIONS

- How do you evaluate numerical expressions with bases of 10 and exponents less than zero?
- How does place value relate to numerical expressions with bases of 10 and exponents less than zero?

LESSON NARRATIVE

In this lesson, students apply the laws of exponents to evaluate numerical expressions with bases of 10 and exponents less than zero. This lesson builds on students' prior knowledge and skills applying the laws of exponents evaluate base 10 expressions with natural number exponents.

COMPONENT SUMMARY

3.3.1 WARM-UP *(~5 MIN.)*

Notice and wonder about bowling ball pyramid

3.3.2 EXPLORATION *(5 MIN.)*

Exploration of powers of 10

3.3.3 TRY IT *(10 MIN.)*

Guess the values of numerical expressions

3.3.4 EXPLORATION *(5 MIN.)*

Use calculator to check value of numerical expression

3.3.5 GUIDED INSTRUCTION *(10 MIN.)*

Apply laws of exponents to evaluate numerical expressions

3.3.6 WRAP-UP *(~5 MIN.)*

Reflect on when they felt proud and confused

RESOURCES

GLOSSARY

Key Terms:

- Negative Powers of 10

Additional Terms:

- N/A

MATERIALS

- Calculators
- (Optional) Highlighters or Colored Pencils
- (Optional) Scissors

ADDITIONAL PREPARATION

- N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may incorrectly think the values of expressions with negative exponents must be negative.
- Students may incorrectly think that a number with an exponent of zero equals zero.
- Students may believe that they can apply a law of exponents to addition and subtraction expressions.

THINGS TO CONSIDER

- Consider displaying the laws of exponents and adding examples with bases of 10 and negative exponents.
- Consider reviewing the names of place values of the decimal representations of negative powers of 10 or providing them on an anchor chart.

LITERACY AND LANGUAGE ROUTINES

Notice and Wonder

In 3.3.1 Warm-Up, prompt students to reflect on what they notice and wonder about a photograph of a pyramid of bowling balls. Call on a variety of students to share what they noticed and wondered. This activity provides a low floor, as all students can participate, and a high ceiling, as students can be challenged to think of multiple noticing and wonderings.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.4.1 Discussion and Reflection

In 3.3.4 Exploration, prompt students to discuss if their guesses from 3.3.3 Try It match the values obtained using a calculator. Next, ask students to consider which expressions could be evaluated using the laws of exponents. Discussions of how the laws of exponents can be applied to the expressions will help build student confidence and understanding.

DIFFERENTIATE INSTRUCTION

To provide a visual entry point in 3.3.4 Exploration, prompt students to highlight how the law of exponents impacted each expression. To provide a kinesthetic entry point, consider structuring question 2 as a matching activity, where students cut the cells in the table out and practice matching the expressions to the laws of exponents.

Mathematical Statement	Value Written Using a Power of 10	Law of Exponents
$10^{-5}(10^{8})$	10^{3}	*product of powers*

3.3.1 WARM-UP: NOTICE AND WONDER

TEACHER MOVES

Notice and Wonder

Prompt students to reflect on what they notice and wonder about the photograph. Call on a variety of students to share what they noticed and wondered. This activity provides a low floor, as all students can participate, and a high ceiling, as students can be challenged to think of multiple things they may notice and wonder.

3.3.1 Warm-Up: Notice and Wonder

A picture of a bowling-ball pyramid is shown.

a. What do you wonder?
Answers will vary. How many bowling balls are there? How was the pyramid made?

b. What do you notice?
Answers will vary. The bowling balls are different colors. Each row has the same number of bowling balls.

c. Write a question about the pyramid that a 5th grader could answer.
How many bowling balls are in the pyramid?

3.3.2 Exploration: Powers of Ten

Use a scientific calculator to find the decimal value of each. Then, write the equivalent fractional value.

Power of 10	Decimal Value	Fractional Value
10^0	1.0	$\frac{1}{1}$
10^{-1}	0.1	$\frac{1}{10}$
10^{-2}	0.01	$\frac{1}{100}$
10^{-3}	0.001	$\frac{1}{1{,}000}$
10^{-4}	0.0001	$\frac{1}{10{,}000}$
10^{-5}	0.00001	$\frac{1}{100{,}000}$

What do you notice about the relationship between the power of 10 and its value?

The power of ten is the same as the number of places in the decimal value and as the number of zeros in the fractional value.

Work with your partner to complete the table.

Power of 10	Fractional Value	Place Value of 1
10^0	1	Ones
10^{-1}	$\frac{1}{10}$	Tenths
10^{-2}	$\frac{1}{100}$	*Hundredths*
10^{-3}	$\frac{1}{1{,}000}$	*Thousandths*
10^{-4}	$\frac{1}{10{,}000}$	*Ten Thousandths*
10^{-5}	$\frac{1}{100{,}000}$	*Hundred Thousandths*

the power of ten relates to place value.

The exponent is always one less than the total number of places in the place value.

Based on the pattern in the table, what is the value of 10^{-12}, written as a fraction?

$\frac{1}{1{,}000{,}000{,}000{,}000}$

Negative Powers of 10

A base 10 with a negative exponent can be rewritten as a fraction.

$$10^{-a} = \frac{1}{10^a}$$

The value of the fraction relates directly to the place value of the 1 when the quantity is expressed in its equivalent decimal form.

3.3.2 EXPLORATION: POWERS OF TEN

DIFFERENTIATE INSTRUCTION

Consider modeling how to enter negative exponents in calculators for students. Having all students use the same make and model calculator will eliminate potential confusion about how to enter the expressions. For example, you could use the Desmos scientific calculator.

QUESTIONING

How do you determine the equivalent fractional value? How did you determine the place value of the number?

TEACHER MOVES

Consider adding this definition to a classroom Word Wall for students to reference. It is not necessary to extend the conversation beyond negative exponents with a base of 10.

Students will explore how to evaluate expressions with bases that include rational numbers and algebraic expressions in Unit 4.

3.3.3 TRY IT: MAKE A GUESS

ENRICHMENT

How did you determine the value? How did the exponents change? How is the power of 10 related to the value?

DIFFERENTIATE INSTRUCTION

Consider providing one-on-one support for students who rated their confidence as a five or less. Another option is to pair students with higher confidence ratings with students with lower confidence ratings. .

3.3.3 Try It: Make a Guess

Without your partner or a calculator, make a guess on the value of each mathematical statement. Do not worry about being wrong. Remember, mistakes are a part of learning.

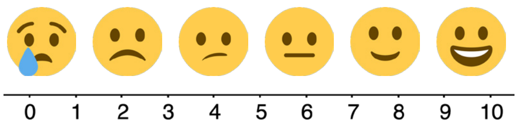
Whenever possible, write the answer as using a power of 10. Show your work or explain your thinking.

Mathematical Statement	My Guess	My Work or Explanation
$10^{-2} + 10^{-1}$	0.11	$0.01 + 0.1$
$10^{-3} \times 10^{-8}$	10^{-11}	0.001×0.000000001
$10^{-1} - 10^{-3}$	0.099	$0.1 - 0.001$
$\frac{1}{10^{-4}}$	$10{,}000$	10^{4}
$(10^{-6})^{2}$	10^{-12}	$10^{-6} \times 10^{2}$
$\frac{10^{-5}}{10^{-7}}$	10^{2}	$10^{-5} - 10^{-7}$

How confident are you in your guesses?

Use the emoji scale to rate your confidence in your guesses.

Answers will vary.



3.3.4 Exploration: Using a Calculator to Check

Use a calculator to complete the following.

a. Find the value of each mathematical statement. Then, if possible, rewrite the value as a power of 10.

Mathematical Statement	Value Found Using the Calculator	Value Written as a Power of 10
$10^{-2} + 10^{-1}$	0.11	*not possible*
$10^{-3} \times 10^{-8}$	0.00000000001	10^{-11}
$10^{-1} - 10^{-3}$	0.099	*not possible*
$\frac{1}{10^{-4}}$	10,000	10^{4}
$(10^{-6})^{2}$	0.000000000001	10^{-12}
$\frac{10^{-5}}{10^{-7}}$	100	10^{2}

b. Compare your earlier guesses to the values you found for each expression, using a calculator.

c. Discuss with your partner which mathematical statements could be rewritten using a power of 10 and which could not be. Summarize your findings.
The addition and subtraction statements could not be rewritten using a power of 10. All the other statements could be rewritten using a power of 10.

Use your findings to name the law of exponents used to rewrite each mathematical statement.

Mathematical Statement	Value Written as a Power of 10	Law of Exponents
$10^{-5}(10^{8})$	10^{3}	*product of powers*
$\frac{10^{-11}}{10^{-4}}$	10^{-7}	*quotient of powers*
$\frac{1}{10^{-10}}$	10^{10}	*quotient of powers*
$(10^{-2})^{3}$	10^{-6}	*power of a power*
$10^{-5} \times 10^{-9}$	10^{-14}	*product of powers*
$\frac{10^{-3}}{10^{-7}}$	10^{4}	*quotient of powers*
$(10^{-6})^{-2}$	10^{12}	*power of a power*

3.3.4 EXPLORATION: USING A CALCULATOR TO CHECK

TEACHER MOVES

Discussion and Reflection

Prompt students to discuss if their guesses from 3.3.3 Try It match the values obtained using a calculator. Note that depending on the calculator used, the calculator may report the answer in scientific notation. If using the Desmos scientific calculator, there is an option to convert it to a fraction so that students can see the value that way. Next, ask students to consider which expressions could be evaluated using the laws of exponents. Discussions of how the laws of exponents can be applied to the expressions will help build student confidence and understanding.

ENRICHMENT

Why is it not possible to write some statements using a power of 10? What do you notice about the operations in the expressions where the value cannot be written using a power of 10?

TEACHER MOVES

Look for students who may not have connected that there were no rules for addition and subtraction in Lesson 1 so there would be no rules in this lesson. Students often confuse the action of adding exponents when multiplying and subtracting exponents when dividing to adding and subtracting numeric exponential expressions.

DIFFERENTIATE INSTRUCTION

Provide a visual entry point by prompting students to highlight how the law of exponents impacted each expression. To provide a kinesthetic entry point consider structuring question 2 as a matching activity, where students cut the cells in the table out and practice matching the expressions to the laws of exponents.

3.3.5 GUIDED INSTRUCTION: APPLYING THE LAWS OF EXPONENTS

TEACHER MOVES

Prompt students to discuss which law(s) of exponents they applied. Ask students to discuss if they agree or disagree and why.

3.3.5 Guided Instruction: Applying the Laws of Exponents

ıe Laws of Exponents can help when rewriting athematical statements as a power of 10. Sometimes ore than one law can be applied.

the table from the Exploration, $\frac{10^{-11}}{10^{-4}}$ was written as 10^{-7}. ıother way to write 10^{-7} is $\frac{1}{10^7}$. Each of these expressions present the same value.

Write each mathematical statement in an equivalent form by applying one or more laws of exponents. Then, write the equivalent decimal value.

Mathematical Statement	Value Written as a Power of 10	Decimal Equivalent
$\left(\frac{1}{10^{-4}}\right)^2$	$\frac{1}{10^{-8}}$	100,000,000.0
$(10^{-11})^3 \times 10^{27}$	10^{-6}	0.000001
$\frac{10^{-13}}{10^{-5}} \cdot 10^8$	10^0	1.0

ımbers like one-billionth, also known as a nano, can be ritten using powers of 10.

Write one-billionth, or $\frac{1}{1,000,000,000}$, as a power of 10.

10^{-9}

One septillionth, $0.000000000000000000000001$, is also known as yocto. Write one septillionth as a power of 10.

10^{-24}

3.3.6 Wrap-Up: Reflection

Complete the table as you reflect on today's lesson.

Today I deserve a  for …	*Answers will vary.*
Today I was feeling  about …	*Answers will vary.*

3.3.6 WRAP-UP: REFLECTION

DIFFERENTIATE INSTRUCTION

Consider labeling each picture to ensure that each and every student correctly interprets the prompts.